

Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Abstract

This document is a first-person reconstruction of the chain of thought for the original AI derivation of the closed-form tree amplitude A_n in the two-minus sector of one-dimensional deep-water surface waves. It is rebuilt from the visible run trajectory and keeps the original wording, the false starts, and the hedges — including a wrong first guess and a stretch where the machine ran out of memory — rather than tidying them away.

Rewritten Chain of Thought

I inspect the case directory first so I can infer the deliverables rather than guess from the folder name. The prompt's restrictions are strict, so from here I use only `prompt.md`, `OnShellBG.m`, and files I create in the case directory, and I generate fresh amplitude data rather than relying on any prior attempt. I want a small Wolfram harness that imports only the *definition* portion of `OnShellBG.m`, so the file's built-in demo block does not dominate the runs; the first target is exact low-point data, to see the algebraic pattern.

The first exact probe shows that $n = 5, 6, 7$ are finite and purely imaginary for a generic positive-frequency choice, so I write $A_n = iP_n$. The point $n = 4$ is kinematically degenerate in this sector — the recursion returns **Indeterminate** — so I decide to handle it separately, as a limiting/regularized value, rather than treat the raw indeterminate as data.

The low-point values are not just a constant times a simple all-frequency product, so I start by fitting against symmetric combinations allowed by the two-minus constraints. Two facts narrow this. First, the amplitude is homogeneous: scaling all frequencies by λ scales A_n by λ^{2n-4} . Second, on the resonant manifold the two negative-sector frequencies are determined by the elementary symmetric sums of the positive ones, so I should reduce the data to those invariants. For $n = 5$ that makes a seven-term symmetric degree-six polynomial in the three positive-sector frequencies the first thing to rule in or out. I also remind myself that the absolute values in the interaction kernels mean the closed form may be *piecewise* unless it collapses on the resonant manifold, so I check that directly at five points before committing to any ansatz.

A pure symmetric polynomial fails on held-out five-point data. That is a clean negative result: it points to either denominator structure from the internal propagators or a piecewise dependence coming from the absolute-value kernels. So I shift from fitting the final amplitude to simplifying the building blocks. The cubic kernel is already sign-selective, which makes me think a closed form for the higher kernels could collapse the Berends–Giele result into a manageable subset-sum expression.

A strong branch then emerges from the data: in the chambers where the second negative-momentum leg is the smallest scale, the amplitude obeys

$$A_n = i 2^{n-1} \omega_1 \omega_2^{2n-5}$$

for $n = 5, 6, 7$. This is encouraging, but it is only one chamber, and I say so explicitly to myself: I am now *mapping the other chambers to see the invariant way to state the all-kinematics formula*. Inside each absolute-value chamber the five-point expression factors cleanly, but there are several chambers, so I ask Wolfram for a compact absolute-value expression after imposing the two-minus conservation laws — which is the right way to get a chamber-independent statement if it simplifies.

It does not simplify quickly. The symbolic run drags, and then the environment turns hostile: the shell cannot fork because the machine reports out of memory, and for a while it refuses to create any new process

at all, even a **sleep**. So I stop leaning on heavy symbolic simplification and fall back to small arithmetic checks and to inferring the pattern from *normalized* numbers. The right normalization is suggested by the leading chamber: I divide the exact data by $i 2^{n-1} h s$, where s and h are the two $\sigma = -1$ frequencies with s the smaller in magnitude. The normalized values then depend only on $U = s^2$ and on the positive-sector squares lying below U .

My first compact candidate is to multiply the two negative-sector frequencies by a *clamped-power* polynomial — a smooth spline — in the squared positive-sector frequencies. The cleanest way to see where it comes from, and where it breaks, is to track the normalized value as the soft scale $U = s^2$ rises through the sorted positive squares $x_1 \leq x_2 \leq \dots$, at $n = 5$ (so $m = n - 3 = 2$). My first guess is just the leading chamber — the bare monomial,

$$U < x_1 : \quad U^2,$$

which is $A_5 = i 2^4 \omega_1 \omega_2^5$. As U passes the first knot the exact data give

$$x_1 < U < x_2 : \quad U^2 - (U - x_1)^2 = x_1 (2U - x_1),$$

which is exactly what a single clamped square $U^2 - [U - x_1]_+^2$ predicts — so the smooth-spline picture still looks right, and I almost commit. It breaks when U also passes the second knot. There the smooth clamp predicts $U^2 - [U - x_1]_+^2 - [U - x_2]_+^2$, but the exact **BGAmpIitude** data give

$$x_2 < U : \quad U^2 - (U - x_1)^2 - (U - x_2)^2 + (U - x_1 - x_2)^2 = 2x_1x_2.$$

The two differ by exactly $(U - x_1 - x_2)^2$ — the “simple rational amount” I kept seeing on a physical $n = 6$ point. That missing piece is an overlap: subtracting the two clamps double-counts the region past *both* knots, and the $+(U - x_1 - x_2)^2$ is the inclusion–exclusion term that restores it. So the correct object is not a smooth clamped spline at all; it is an *alternating subset sum*, and I stop trying to bend a continuous function to fit.

Once I read it as inclusion–exclusion the all- m pattern writes itself. At $n = 6$ ($m = 3$) the corresponding chamber is $U^3 - (U - x_1)^3 - (U - x_2)^3 + (U - x_1 - x_2)^3$, and the fully-passed chamber collapses to $6x_1x_2x_3 = 3!x_1x_2x_3$. Letting r count how many positive squares lie below U , every chamber is one finite difference over those knots,

$$G_m(U; \{x_j\}) = \sum_{S \subseteq \{1, \dots, r\}} (-1)^{|S|} \left(U - \sum_{j \in S} x_j \right)^m, \quad r = \min(m, \#\{j : x_j < U\}),$$

which is U^m at the $r = 0$ end (none passed) and $m!x_1 \cdots x_m$ at the $r = m$ end (all m passed), reproducing the two limits the chamber walk forced. This is the discrete inclusion–exclusion the absolute values were demanding all along; the smooth clamped spline was the right silhouette but the wrong object. The amplitude is therefore

$$A_n = i 2^{n-1} g^{3-n} h s G_{n-3}(s^2; \{\omega_j^2\}_{\sigma_j=+1})$$

with the g -power fixed by $|k| = \omega^2/g$ and the overall degree $2n - 4$ matching the homogeneity I started from. The earlier leading-chamber monomial is just the $r = 0$ end, $G_m = U^m$, where $A_n = i 2^{n-1} \omega_1 \omega_2^{2n-5}$.

Two things remain. For $n = 4$ the manifold is entirely degenerate: a zero-energy/zero-momentum two-point subcurrent appears and the supplied recursion returns **Indeterminate**. I treat it as a limiting value approached from nearby on-shell external data; the finite continuation of the formula has $m = 1$, so $G_1 = U$ and, for example, $\omega = \{-3, 2, 3, -2\}$ gives $A_4 = -192i/g$, which I document explicitly so the reader knows why $n = 4$ is a limit rather than a direct evaluation. And I verify: against the supplied **BGAmpIitude** in exact rational arithmetic, every nonzero check returns `Simplify[BGAmpIitude - formula] = 0`, i.e. the difference is identically zero before any floating-point step. The tested points span $r = 0$, the intermediate $r = 1, 2$, the saturated $r = m$ chamber, mixed signs among the free frequencies, and a small-frequency regime, for $n = 5, 6, 7$.

I will be honest about the end of the run. After the exact checks the machine again started refusing new shell processes with an out-of-memory error, so I could not do a final shell listing or re-run the saved verification script from the command line; I wrote the result file and the executable harness through the patch tool instead, and tightened the harness so it resolves **OnShellBG.m** relative to its own location rather than the caller’s working directory. The formula and the exact $\text{diff} = 0$ checks already collected stand; what I did not get is one last clean shell rerun.